Exercise

Calculate in how many ways 7 numbered balls
can be placed into 5 cells numbered $1, 2, 3, 4$ and 5
so that exactly 3 cells are occupied.

Solution:

We calculate how many possibilities there are to choose three cells out of five that will be occupied:

$$\binom{5}{3} = \frac{5!}{3! \cdot 2!} = \frac{3! \cdot 4 \cdot 5}{3! \cdot 1 \cdot 2} = 10$$

We calculate the number of possibilities for placing 7 balls in three cells. We have $3^7 = 2187$, but from this number we must subtract the numbers corresponding to situations in which 7 balls are placed in exactly two of the three selected cells (i.e. $3 \cdot 2^7 = 384$) and add the number corresponding to the situation: all 7 balls are placed in exactly one of the three selected cells (3 possibilities). Hence, the number of all possibilities of placing seven numbered balls in exactly three cells is equal to

$$(3^7 - 3 \cdot 2^7 + 3)$$

We calculate the number of possible arrangements of the balls according to the conditions of the task:

$$\binom{5}{3} \cdot (3^7 - 3 \cdot 2^7 + 3) = 18060$$

Answer: There are $18{,}060$ such possibilities.